

System Card

Face Verification Pipeline — v1.0-final
Course: MSML/MSAI 605 | Milestone 4 | 2025

1. System Overview

This system is a face verification pipeline that accepts two face images and returns a binary decision — SAME (same person) or DIFFERENT (different people) — along with a calibrated confidence score and per-stage latency measurements.

Pipeline:

Stage	Description
Input	Two RGB face images (JPEG, PNG, etc.)
Preprocessing	Resize to 160x160; per-image mean/std normalization
Embedding	FaceNet InceptionResnetV1 pretrained on VGGFace2 (512-dim)
Similarity	Cosine similarity between embeddings; score in [-1, 1]
Decision	score >= 0.35 → SAME (1), else DIFFERENT (0)
Confidence	sigmoid(10 x (score - 0.35)); range [0, 1]; 0.5 = boundary

Output Fields:

Field	Type	Description
score	float [-1,1]	Cosine similarity between embeddings
decision	int {0,1}	1=SAME, 0=DIFFERENT
confidence	float [0,1]	Calibrated confidence; 0.5 = boundary
threshold	float	Operating threshold (0.35)
latency_total_s	float	End-to-end wall time (seconds)
latency_preprocess_s	float	Preprocessing stage time
latency_embedding_s	float	Embedding stage time
latency_scoring_s	float	Scoring stage time

2. Intended Use

Supported use cases:

- Academic research and coursework: evaluating face verification algorithms with oversight.
- Offline batch verification: processing image pairs where human review follows.
- System benchmarking: comparing embedding-based vs. pixel-based verification baselines.
- Educational demonstrations: understanding threshold calibration, confidence scoring, and latency profiling.

Out-of-scope uses (explicitly not supported):

- Real-time surveillance or tracking of individuals in any space.
- Access control, authentication, or security systems without independent safety review.
- Any application affecting legal rights, employment, housing, or law enforcement.
- High-stakes identity verification in production environments (border control, financial).
- Deployment on populations substantially different from the LFW distribution without re-evaluation.
- Use on minors or in contexts involving children without ethical review and consent.

3. Data Summary

Property	Value
Dataset	Labeled Faces in the Wild (LFW) via TensorFlow Datasets
Split policy	70% train / 15% val / 15% test — by identity (no leakage)
Pairs per split	500 positive + 500 negative (deterministic, fixed seed)
Image format	RGB, 250x250 native; resized to 160x160 in preprocessing

Key data limitations:

- Celebrity/public figure bias: LFW consists largely of news-media images. The demographic distribution does not represent the general population.
- No demographic metadata: Ground-truth demographic labels are unavailable. Quantitative subgroup fairness analysis is not possible.
- Class imbalance by identity: Some identities have many more images; an optional cap (`max_images_per_identity`) was introduced in Milestone 2 to reduce dominance.
- Image quality variation: Lighting, pose, resolution, and occlusion vary. No face alignment is performed.

4. Operating Threshold and Key Metrics

Parameter	Value
Operating threshold	0.35
Selection rule	Max balanced accuracy (sweep of 100 candidate thresholds)
Selected on	Validation split
Config location	<code>configs/m3.yaml</code> → <code>inference.threshold</code>

Full metric tables (accuracy, balanced accuracy, precision, recall, F1, TPR, FPR, confusion matrix) are recorded in `outputs/eval/val/metrics.json` and `outputs/eval/test/metrics.json`. To reproduce: follow `reports/reproducibility_checklist.md`.

Confidence interpretation:

Confidence	Meaning
1.0	Very high same-person confidence (score far above threshold)
> 0.5	Same-person lean
0.5	Exactly on the decision boundary
< 0.5	Different-person lean
0.0	Very high different-person confidence (score far below threshold)

5. Failure Modes and Limitations

Low-quality or non-frontal images: No face detection or alignment is performed. Occluded, profile, or very low resolution images produce degraded embeddings and unreliable scores.

Near-boundary scores: Pairs with scores within +/-0.05 of the threshold produce confidence values near 0.5. These decisions are inherently uncertain and should not be treated as definitive.

Visually similar different-identity pairs: The system can produce false positives when two different people are visually similar (e.g., twins, family members).

Non-face inputs: The pipeline has no face detector. Non-face images are embedded and compared without warning, producing meaningless scores.

Model fallback degradation: If facenet-pytorch is unavailable, the pipeline falls back to TF Hub MobileNetV2 or normalized pixels. Threshold 0.35 is only calibrated for FaceNet; using a fallback with the same threshold will degrade performance.

Distribution shift: Performance may degrade on images from substantially different distributions (different cameras, populations, forensic contexts).

6. Fairness-Related Risks and Misuse Concerns

Important caveat: LFW does not include verified demographic labels. The analysis below is qualitative. No subgroup performance numbers are claimed because the data does not reliably support them.

Demographic representation imbalance: LFW is known to underrepresent certain demographic groups (particularly women and some ethnic backgrounds). A threshold calibrated on this distribution may generalize less reliably to underrepresented groups.

Image quality disparities: If certain populations are more likely to be photographed under poor conditions, the system may perform less reliably for those groups — not by design, but by calibration gap.

Surveillance and tracking misuse: Face verification systems can be misused for surveillance or unauthorized identification. This system is not validated for any such use.

False positive harms: In any high-stakes deployment (explicitly out of scope), false positives — incorrectly matching different people — could cause serious harm. Error analysis artifacts document these cases.

False negative harms: False negatives — failing to match the same person — could also cause harm in access control settings. Both error types must be considered before any deployment.

Recommended mitigations for any future deployment:

- Evaluate on a demographically representative held-out dataset before any deployment.
- Use human review for near-boundary decisions (confidence near 0.5).
- Clearly document intended use and prohibit deployment for surveillance or legal purposes.
- Re-calibrate the threshold if the deployment distribution differs from LFW.

7. Operational Constraints

Constraint	Details
Input format	Any RGB image readable by Pillow (JPEG, PNG, etc.)
Threshold validity	0.35 calibrated for FaceNet backend only
Model fallback	FaceNet -> TF Hub MobileNetV2 -> normalized pixel vector
CPU latency	Embedding stage dominates; see reports/profiling_report.pdf
GPU support	Optional; torch.cuda auto-detected
Concurrency	Tested: ProcessPoolExecutor, 4 workers, 50 requests
Container	Docker image face-verifier:v1.0-final
Python version	3.10+
Key deps	See requirements.txt

8. Reproducibility Pointer

Item	Location
How-to-run instructions	README.md
Exact reproduction commands	reports/reproducibility_checklist.md
Final inference config	configs/m3.yaml
Profiling report	reports/profiling_report.pdf
Final Git tag	v1.0-final
Docker entrypoint	scripts/cli_infer.py via Dockerfile
Test suite	pytest tests/ -v

A grader can reproduce the core outputs by following the reproducibility checklist. No LFW data download is required to verify CLI behavior or run the test suite (synthetic test images are generated by `make_test_imgs.py`).